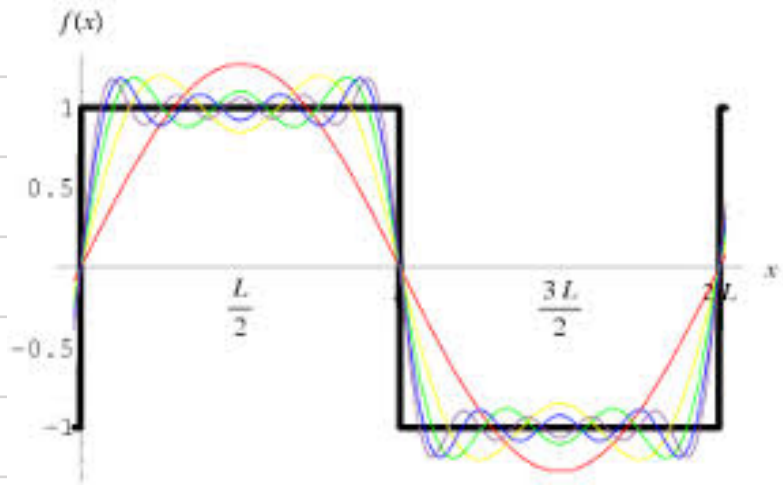
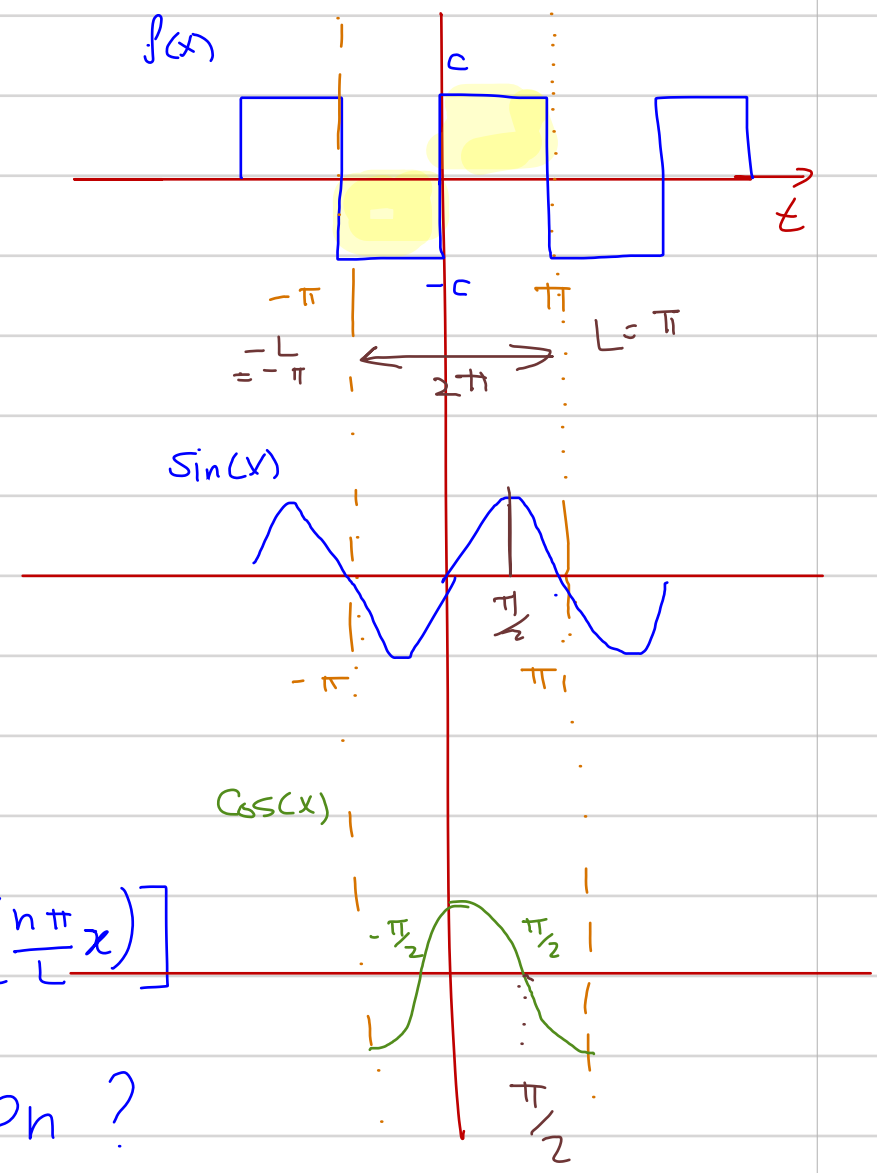
Physical Layer

f(x)





Fourier Series

$$f(x) = a_0 + \sum_{n=1}^{N}\left[a_n \cos\left(\frac{n\pi}{L}x\right) + b_n \sin\left(\frac{n\pi}{L}x\right)\right]$$

fourier found a_0, a_n, & b_n ?

$$a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx$$ when the period $P = 2L$

P

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi}{L}x\right)dx$$

$\frac{2}{P}$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi}{L}x\right)dx$$

$\frac{2}{P}$

$n = 1, 2, 3, \ldots$

find a_0, a_n, b_n ?

$L =$

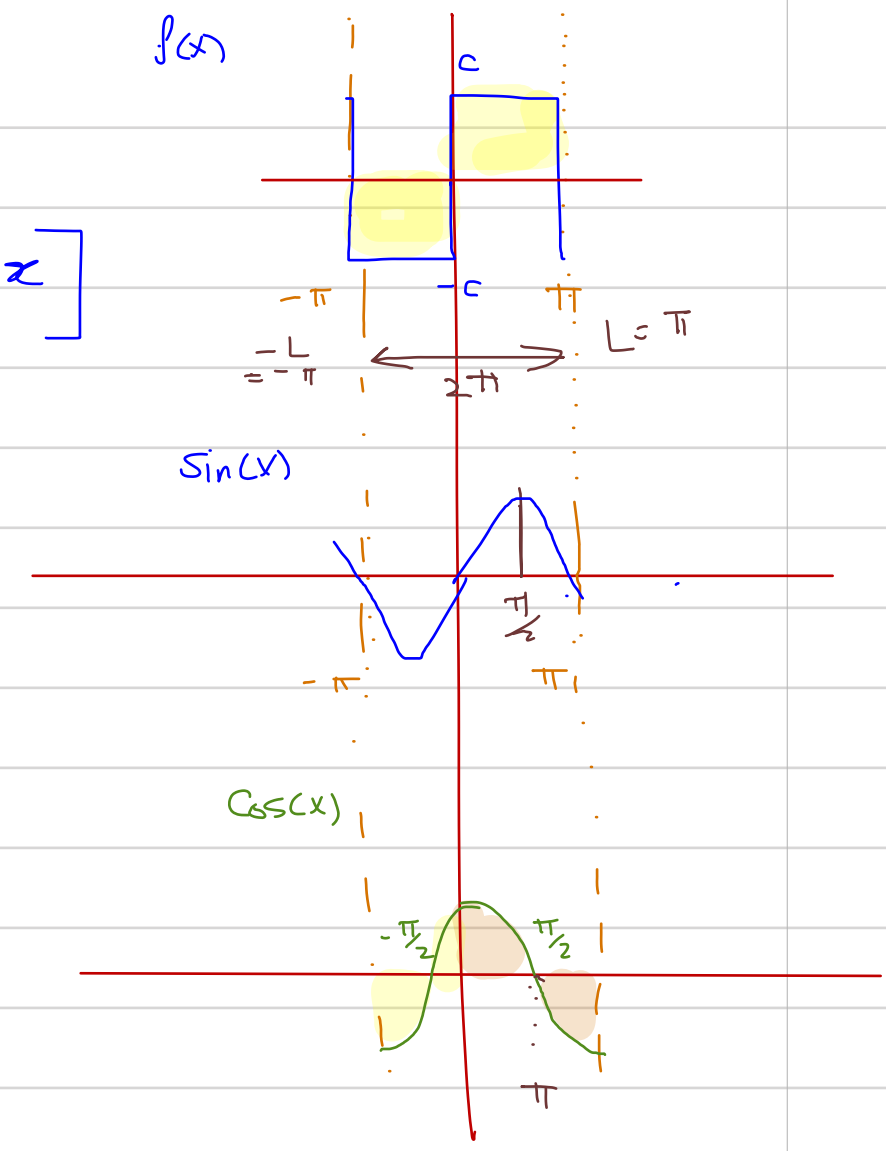
$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{2\pi}\left[\int_{-\pi}^{0}(-c)\,dx + \int_{0}^{\pi} c\,dx\right] = 0$$

$$a_1 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos\left(\cancel{\frac{\pi}{\pi}}^{\,1} x\right)dx =$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} -c\cos(x)\,dx + \int_{0}^{\pi} c\cos(x)\,dx\right]$$

$$= \frac{1}{\pi}\left[-c\int_{-\pi}^{0}\cos(x)\,dx + c\int_{0}^{\pi}\cos(x)\,dx\right]$$

$$= \frac{1}{\pi}\left[-c\times 0 + c\times 0\right] = 0$$



$$b_1 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin\left(\cancel{\frac{\pi}{\pi}}^{\,1} x\right)dx$$

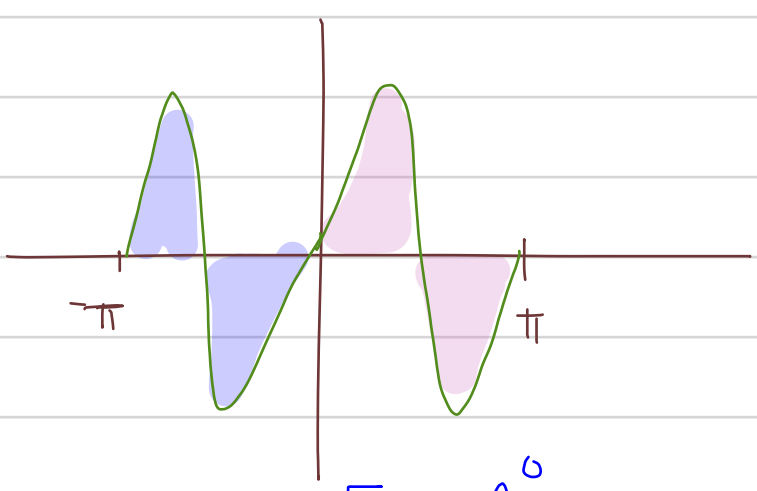
$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} -c\sin(x)\,dx + \int_{0}^{\pi} (c)\sin(x)\,dx\right]$$

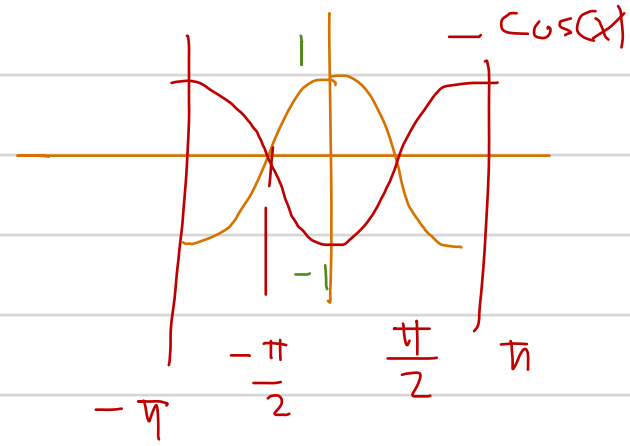
$$= \frac{1}{\pi}\left[(-c)(-2) + (c)(2)\right]$$

$$= \frac{4c}{\pi}$$

$$\int \sin(x)\,dx = -\cos(x)\Big]_{-\pi}^{0}^{\pi} = -2 \text{ or } 2$$

$$b_2 = \frac{1}{\pi}\left[\int_{-\pi}^{0} -c\sin(2x)\,dx + \int_{0}^{\pi} c\sin(2x)\,dx\right] = 0$$





$$b_3 = \frac{1}{\pi}\left[\int_{-\pi}^{0} (-c)\sin(3x)\,dx + \int_{0}^{\pi} (c)\sin(3x)\,dx\right]$$

$$= \frac{1}{\pi}\left[(-c)\left(\frac{-2}{3}\right) + c\left(\frac{2}{3}\right)\right] = \frac{4c}{3\pi}$$

$b_4 = 0$, $b_6 = 0$, $b_8 = 0 \ldots$

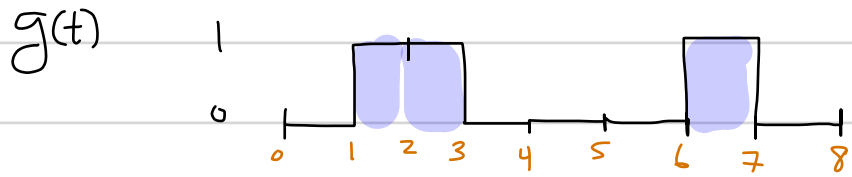
$$b_5 = \frac{4c}{5\pi}, \quad b_7 = \frac{4c}{7\pi}$$

f(x) is ?

$$f(x) = \frac{4c}{\pi}\sin(x) + \frac{4c}{3\pi}\sin(3x) + \frac{4c}{5\pi}\sin(5x) + \frac{4c}{7\pi}\sin(7x) \ldots\ldots$$

Example

8 bit ASCII "b" = 0 1 1 0 0 0 1 0



$P = T = 8$

Find the fourier series representation of 'b'?

$$a_0 = \frac{1}{P}\int_0^T g(t)\,dt = \frac{1}{8}\int_1^3 1\,dt + \int_6^7 1\,dt$$

$$\frac{1}{8}(2 + 1) = 3/8$$

$$a_n = \frac{\cancel{2}^{1}}{\cancel{8}_{4}\,\cancel{P}}\int_0^T g(t)\cos\left(\frac{n\pi\,\cancel{2}^{1}}{\cancel{8}_{4}\,\cancel{P}}t\right)dt = \frac{2}{8}\left(\int_1^3 \cos\left(\frac{\cancel{2}^{1}\pi n t}{\cancel{8}_{4}}\right)dt + \int_6^7 \cos\left(\frac{\cancel{2}^{1}\pi n t}{\cancel{8}_{4}}\right)dt\right)$$

$$= \frac{1}{4}\left(\left[\frac{\sin(\pi n t/4)}{\pi n/4}\right]_1^3 + \left[\frac{\sin(\pi n t/4)}{\pi n/4}\right]_6^7\right)$$

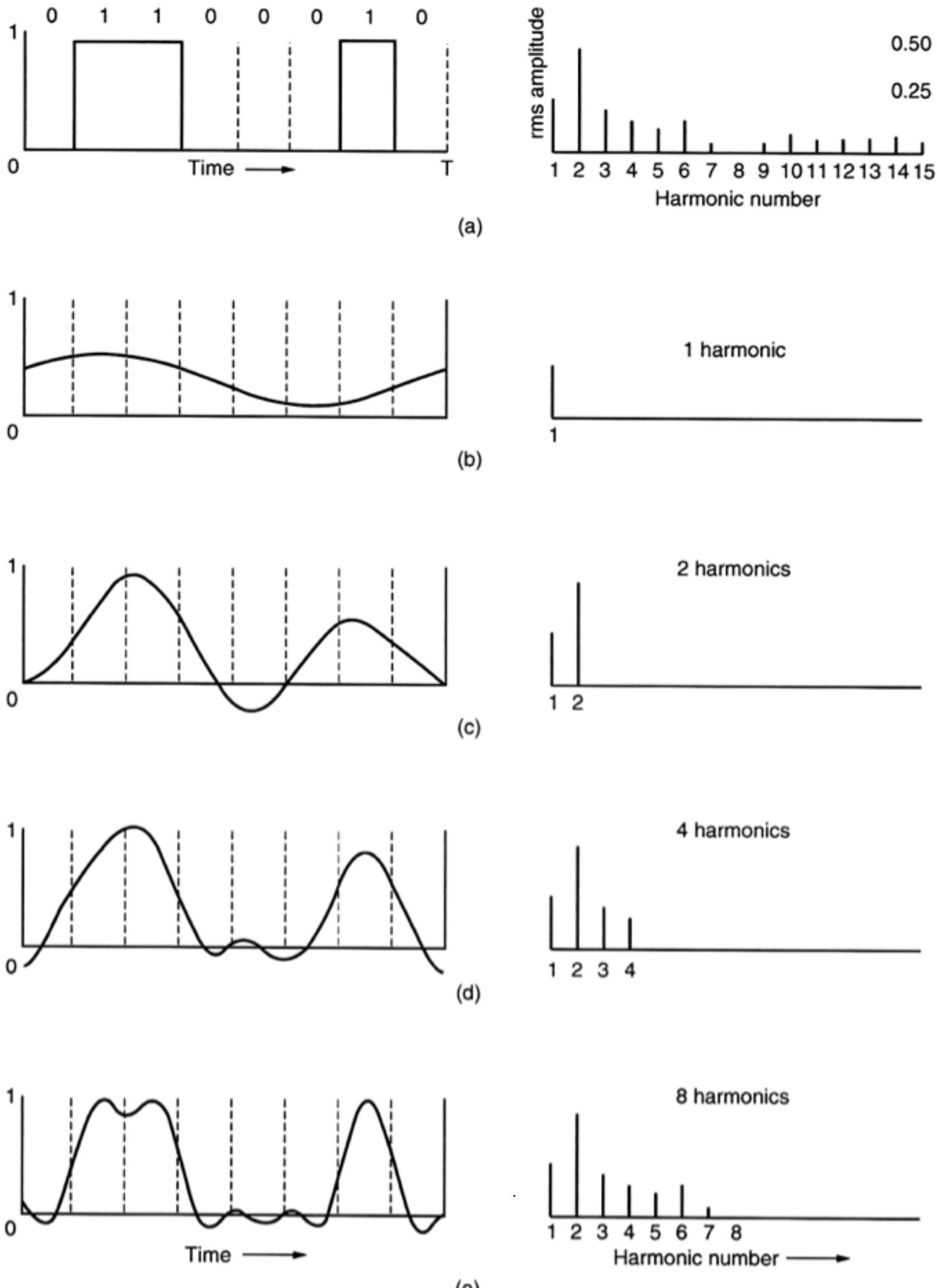
$$= \frac{1}{\cancel{4}}\cdot\frac{1}{\pi n/4}\left[\sin\left(\frac{3\pi n}{4}\right) - \sin\left(\frac{\pi n}{4}\right) + \sin\left(\frac{7\pi n}{4}\right) - \sin\left(\frac{6\pi n}{4}\right)\right]$$

$$b_n = \frac{1}{\pi n}\left[\cos\left(\frac{\pi n}{4}\right) - \cos\left(\frac{3\pi n}{4}\right) + \cos\left(\frac{6\pi n}{4}\right) - \cos\left(\frac{7\pi n}{4}\right)\right]$$

For different harmonics sub for n.

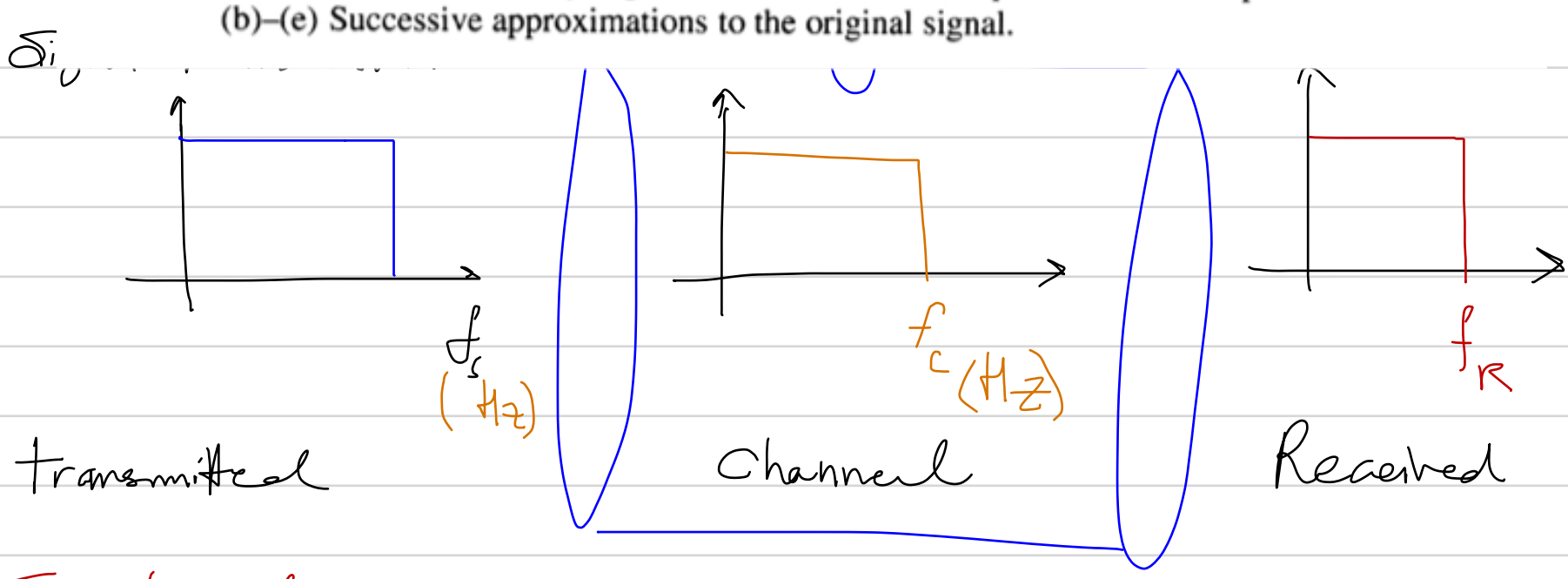
homework plot g(t) vs. it fourier series representation using matlab. Keep increasing n until the plotted series is close to the original signal.

The root mean square amplitudes $\sqrt{a_n^2 + b_n^2}$ is important because their square are proportional to the energy transmitted at the corresponding frequency.



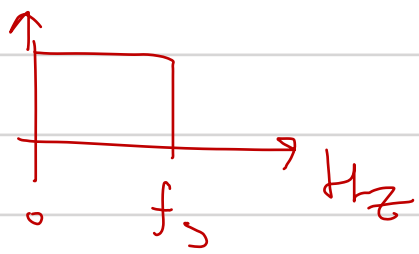
(e)

Figure 2-1. (a) A binary signal and its root-mean-square Fourier amplitudes. (b)–(e) Successive approximations to the original signal.

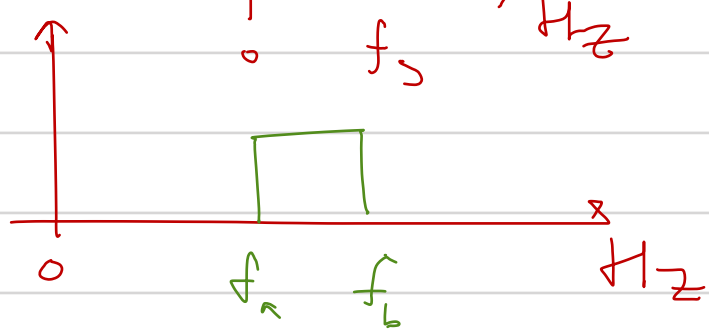


Two types of signals

base band signal



Pass band signal



how is it related to data Rate?

Given a rate of b bits/sec, the time required to send 8 bits is $\overbrace{\frac{8}{b}}^{T}$ sec.

So, the frequency of the first harmonic (min) is $\frac{1}{T}$ which is $\frac{b}{8}$ Hz.

. The voice-graded line has a bandwidth of 3000Hz (3 KHz). This means that the largest number of harmonics I can send through is

$$\frac{3000 \text{ Hz}}{(b/8)}.$$

b

Data Rate (bps)	time to send 8 bits (sec) (T)	First harmonic	max # of harmonics in 3KHz
300 bps	$8/300 = 26.67$	$\frac{300}{8} = 37.5$	$3000/37.5 = 80$

Bps	T (msec)	First harmonic (Hz)	# Harmonics sent
300	26.67	37.5	80
600	13.33	75	40
1200	6.67	150	20
2400	3.33	300	10
4800	1.67	600	5
9600	0.83	1200	2
19200	0.42	2400	1
38400	0.21	4800	0

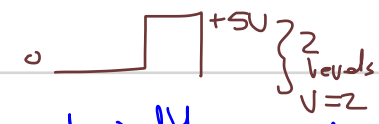
Figure 2-2. Relation between data rate and harmonics for our example.

Nyquist Capacity of noiseless channel: (1924, AT&T)

The maximum data rate $= 2\,B \log_2 V$ bits/sec

bandwidth (Hz)

the number of levels in a signal



The maximum data rate over noiseless channel with 3KHz bandwidth and binary (two levels) signal is 6000 bps.

Shannon Capacity (noisy channel) (1948)

SNR: Signal to noise ratio S/N, measured in dB: deci bels

10 bell

$$10 \log_{10}(SNR)$$

$$1000 = 30 \text{ dB}$$

Shannon's Capacity $\Leftrightarrow$ the maximum data rate that can be sent over a channel with width B Hz is

$$B \log_2(1 + SNR)$$

ADSL, with bandwidth of 1 MHz and SNR of 40 dB the line capacity is $(1 \text{ MHz}) \log_2(1 + 10000) = 13$ Mbps